

Language-Like Efficiency or Efficient Biology? Cetacean Vocalizations Reconsidered

Ådne SKJÆVELAND JØSSING
adne.jossing@student.uni-tuebingen.de

Animal and Plant Communication (WiSe 2025-26)
Seminar für Sprachwissenschaft
Universität Tübingen

Abstract

Youngblood, 2025 and related work have documented Zipf's law of abbreviation and Menzerath's law in cetacean vocalizations, framing these findings as evidence of language-like communication. This paper argues that both laws are signatures of biological efficiency rather than language specifically. They appear in genomes, music, and non-referential animal signals, none of which are language. With Hockett's design features as a practical framework, I identify signal-observable properties that together with the attested laws would constitute stronger evidence for language-likeness: duality of patterning, productive vocabulary growth, displacement, and second-mention reduction. Evaluated against these, current data on cetaceans shows promise, but correlations are mostly untested. This reflects the state of available data more than it is a conclusion about cetacean abilities.

1 Introduction

In recent years interest in the structural properties of cetacean communication has increased significantly. Youngblood, 2025 analyzed vocal sequences from 16 whale species and found that most adhere to Menzerath's law—the tendency for longer sequences to contain shorter elements—with effect sizes comparable to or exceeding those found in human speech. Humpback and blue whales additionally exhibit Zipf's law of abbreviation, the tendency for more frequent elements to be shorter. Youngblood frames this as a sign of “language-like efficiency”. Arnon et al., 2025 identified Zipfian statistical structure in humpback whale song using segmentation methods borrowed from human

infant language acquisition research. Andreas et al., 2022, with the interdisciplinary Project CETI initiative, have proposed applying natural language processing tools directly to sperm whale communication, with the explicit goal of decoding its structure and meaning.

These developments raise a question that is rarely asked directly: To what extent do Zipf's law of abbreviation and Menzerath's law, as observed in cetacean vocalizations, constitute evidence for language-like communication, and what additional conditions would need to be met for stronger claims to be warranted?

This paper argues that Zipf's law of abbreviation and Menzerath's law, while solidly present in cetacean vocalizations and genuinely interesting as findings in comparative biology, are signatures of biological efficiency and compression rather than signatures of language. Both laws appear in biological systems with no communicative function at all: genomes, protein domain organization, and in communicative systems that lack the structural properties we associate with language, including music and non-referential animal signals. Establishing that cetacean communication is organized for compression is meaningful but it is not the same as establishing that it is directly language-like.

The paper proceeds as follows. Section 2 examines what the two laws measure and why they have been associated with linguistic organization. Section 3 goes over a wider range of non-linguistic systems in which the same laws appear, arguing that they reflect efficient hierarchical organization generally, rather than linguistic structure specifically. section 4 identifies a set of additional signal observable properties: duality of patterning, productive vocabulary growth, displacement, and second-mention reduction. These would constitute stronger evidence for language-likeness, and asks what data would be needed to assess them. Section 5 evaluates the current cetacean evidence against this framework. The paper concludes by arguing that the presence of these laws is better understood as evidence of efficient biological organization than as evidence of language-likeness specifically, and outlines what could be demonstrated for stronger claims to be warranted.

2 Zipf's Law of Abbreviation and Menzerath's Law in Human Language and Whale Vocalizations

There are two statistics which repeatedly show up along with claims of language-like structure in cetacean communication: Menzerath's law, and Zipf's law of abbreviation. Both are organizational features shared between human language and the vocal systems

of other species. The two laws differ in what exactly they measure, and the degree to which they can be found in cetacean vocalizations.

2.1 Menzerath's Law

Menzerath's law, formalized by Altmann and Grotjahn, 1980, states that the longer a linguistic construct, the shorter its constituents. It was originally formulated from the observed relationship between word length and syllable duration in German, and has since been documented across multiple levels of linguistic hierarchy: sentences contain shorter words as sentences grow longer, phrases contain shorter syllables as phrases grow longer. It has been documented as a near-universal tendency in human spoken languages (Torre et al., 2019). The standard interpretation of this is one of compression: as the costs of producing a long sequence accumulate, the speaker compensates by reducing the size of each component.

Youngblood, 2025 finds Menzerath's law in 11 of 16 cetacean species examined, spanning both baleen and toothed whales, noting that the effect sizes are sometimes greater than those observed in spoken human languages.

2.2 Zipf's Law of Abbreviation

Zipf's law of abbreviation predicts that more frequently used elements will be shorter in duration. In human languages this can be observed at multiple levels: common words are shorter than rare ones, and common phonemes are shorter than rare ones. The theoretical grounding is more specific than in Menzerath's law. Kanwal et al., 2017 show through experiments that Zipfian abbreviation emerges in miniature artificial languages when speakers are put under pressure to communicate both quickly and accurately. The law does not emerge from pure speed pressure or accuracy pressure alone, it requires both at once. This means that finding Zipf's law of abbreviation in a communication system is evidence that the system is organized around efficiently transmitting variable, meaningful information.

Bentz and Ferrer-i-Cancho, 2016 confirmed how universal this law is across human languages at scale. They found a negative correlation between word frequency and length without exception across 986 languages, and argued that the trend comes from fundamental principles of information transfer, rather than properties specific to certain languages in particular.

Youngblood, 2025 finds Zipf's law of abbreviation in only two of the five cetacean species

for which element types were categorized: humpback whales and blue whales. He notes that assessing this law requires elements to be categorized into discrete types, which is why it could only be tested in a subset of his dataset: species whose vocalizations have been annotated at a level of detail that allows type-frequency counts. Humpback whales exhibit the law both within phrases and in phrases within songs, and the effect size is comparable to that observed in human languages. This is the clearest of the two findings from the perspective of language-likeness, exactly because of the theoretical conditions the law requires: a system where elements vary in predictability and informativeness, and where there is selection pressure to communicate both quickly and accurately.

Arnon et al., 2025 independently identified Zipfian statistical structure in humpback whale song with segmentation methods taken from human infant language acquisition research. Instead of manually annotating call types, they applied statistical learning algorithms designed to find recurring units in continuous speech, and found that their results follow a Zipfian frequency distribution. This is promising, since the method is less dependent on prior human annotation choices than in the approach used by Youngblood, while still achieving a similar result. This serves as additional confirmation that Zipfian structure in humpback song is not only an artefact of how researchers have chosen to segment it.

3 The Laws Beyond Language

The connection between these laws and language is real, but it does not necessarily imply language-likeness directly. Both Menzerath's law and Zipf's law of abbreviation are instances of a broader phenomenon: compression. Ferrer-i-Cancho et al., 2013 argue that this kind of compression, as a general organizational principle in complex systems, is universal in animal behaviour. Semple et al., 2022 extend this more generally to all biological systems, showing that these regularities appear across species, and across functions much wider than communication alone. This means that finding them in cetacean vocalizations, while notable, does not on its own distinguish a linguistic system from any other efficiently organized biological system.

An interesting illustration of this is in genomes. Nikolaou, 2014 showed that Menzerath's law holds in eukaryotic chromosomal organization, with longer chromosomes containing shorter genes, following the same mathematical form as in linguistic data. Similarly, Protein domain architecture has similar hierarchical compression patterns (Shahzad et al., 2015). Neither of these systems communicates anything.

Zipf's rank-frequency law, which is the broader family Zipf's abbreviation belongs to, appears in music (Rohrmeier et al., 2015). Music is organized, hierarchically structured, culturally transmitted, and in some species sexually selected. This overlaps considerably with humpback whale song particularly. Yet music is not generally considered to convey content or show the compositional semantics that characterize language. Youngblood, 2025 acknowledges this parallel explicitly. This comparison goes both ways: if humpback song resembles music statistically, the language-likeness of interpretation requires additional justification beyond the statistical laws alone.

Even within clearly communicative systems, the laws do not show up consistently, and these systems do not necessarily map neatly onto language. Heesen et al., 2019 found linguistic laws in chimpanzee gestural communication, but Safryghin et al., 2022 showed that the same laws do not hold for chimpanzee sexual solicitation gestures. This suggests that the laws depend on the communicative context, and function in ways that are not yet fully understood. The laws are therefore not stable properties of a communication system as a whole, and can apply in some behavioural contexts and not others within the same species. This dependency on context is however informative in itself: if the laws tracked language-like organization, we would expect them to be more uniformly present across a species' communicative repertoire.

Taken together, the distribution of these laws across various systems suggests that they are signs of efficient hierarchical organization in a broad sense, but do not imply linguistic structure specifically. The appropriate question is therefore not whether cetacean communication is language-like in the sense of sharing these statistical properties, but whether it shares the deeper organizational properties that make human language what it is, and which efficient compression alone does not capture.

4 Further Criteria for Language-Likeness

Establishing that a communication system is organized for efficiency is not the same as establishing that it is language-like in any deep sense. Human language is, among other things, an efficient system, but efficiency is a property it shares with genomes, music, and chimpanzee gestural communication. What distinguishes language from these system is a cluster of other structural properties, many of which are observable from the signal itself without requiring access to the internal states of speakers. This section outlines several such properties, and asks for each one if evidence for it exists in cetacean communication, and what kind of data would be needed to assess it.

4.1 Hockett’s Observable Design Features

The inventory of design features by Hockett and Hockett, 1960 is a practically useful starting point, but it must be stated that it is controversial in modern research. It has been criticized as a checklist approach that misses the underlying cognitive structure of language. This is however also the exact reason it is a good starting point for animal communication, since most of its features are defined in terms of signal behaviour rather than internal representation, which in principle makes them testable in non-human species. The features most relevant to cetacean communication are the following.

The duality of patterning could be said to be the most structurally distinctive property of human language. It refers to the existence of two independent levels of organization: a lower level of meaningless sound units (phonemes) that combine according to rules to form a higher level of meaningful units (morphemes). This is what gives human language its generative power: a small inventory of phonemes can combine to produce an effectively unlimited vocabulary. Whether cetacean vocalizations show anything similar is unclear. Andreas et al., 2022 ask this question: whether sperm whale clicks contain sub-coda features that function like phonetic building blocks, and they note the spectral and temporal properties of individual clicks could carry information other than the inter-click interval patterns used to classify coda types. Sharma et al., 2024 found combinatorial structure in sperm whale codas: certain types co-occur in ways that are not random, but combinatorial structure at the level of whole codas is not the same as duality of patterning, which requires a lower level of meaningless units composing into a higher level of meaningful ones.

Productivity—the capacity to produce and understand an unlimited number of novel signals from a finite set of elements—is related to duality of patterning. In practice, measuring productivity in a non-human system requires evidence that new combinations are produced and responded to appropriately, which is difficult to measure without behavioural experiments. This could however be measured by proxy from the available data, through vocabulary growth curves. This connects directly to Heap’s law, which is discussed further below.

Displacement—the ability to communicate about entities and events not in the immediate context— is one of the features which lacks the most clearly from known animal communication systems. For example, vervet monkey alarm calls refer to predator types, but are made only when the referenced predators are present. It is unknown whether cetaceans show any evidence of displacement. Proving it would require correlating vocalizations with behavioural and environmental contexts in ways that the current datasets

do not support. Andreas et al., 2022 identify this as a key problem.

Arbitrariness—the absence of any direct, natural relationship between the form and meaning of signals—requires first knowing what signals mean, which is not established for most species. This therefore remains untested in cetaceans, even though it is a feature in all of human language.

4.2 Heap’s Law: Vocabulary Growth as a Structural Diagnosis

Heap’s law, also known as Herdan’s law, describes the relationship between corpus size and vocabulary size: as a corpus grows, new word types keep appearing, but at a decreasing rate that follows a power law. In human language, this growth is theoretically unlimited: as long as a text keeps growing, new words keep appearing. If a system has a closed set of vocalizations, vocabulary growth would stagnate completely after the text grows past a certain point. Heap’s law could therefore potentially be a possible measure observable in the corpora available for whether cetacean communication systems have open, productive vocabularies, or closed, finite ones.

Being open-ended is a defining property of human lexicons. Bentz et al., 2017 show across more than 1000 languages that vocabulary growth in human language does not plateau, even in very large corpora, since languages can always generate new forms by affixation, compounding and neologisms.

Conceptually this application of Heap’s law is straight-forward, and feasible given that large annotated datasets exist, with distinct call types annotated separately. To my knowledge this analysis has not been systematically applied to cetacean data, though existing datasets could be compatible. The main limitation here is the same one as in Zipf’s law of abbreviation: the elements need to be categorized into discrete types, and the results will vary depending on how type boundaries are drawn.

4.3 Second-Mention Reduction: Predictability and Discourse Structure

Words and phrases in human language are shorter and less phonotactically precise when they have been mentioned recently or are predictable from the context. This is known as the second-mention reduction law, or uniform information density (Jaeger and Levy, 2007, Fowler and Housum, 1987). This is local and context-sensitive: the same element is produced differently depending on whether it is new information or not.

It is an open question whether this exists in cetacean communication. But this, like Heap’s law, is something the available datasets could theoretically be testable against. Finding second-mention reduction in a communication system is significant, because it implies that the system is organized around managing information flow between speakers, and that they in some sense are modeling the knowledge of the listener, and adjusting the signal form accordingly.

4.4 The Measurement Gap

The properties from this section are all in principle measurable from acoustic and behavioural data, without needing a window into the cognitive architecture of cetaceans. This is important, since the question of whether whale communication is language-like should not depend on whether they have Chomskyan syntax or recursive mental representations. These are unanswerable given current data and methodology, and we should be cautious of trying to fit theorized human cognitive frameworks over other species, as Andreas et al., 2022 warn against.

Zipf’s law and Menzerath’s law are documented, but there would be much stronger evidence for language-like organization if the properties outlined in this section could also be documented. They remain largely unattested in cetaceans. The data may be partially compatible, but the methods to test them have not yet been systematically applied. The remainder of this paper asks what the existing cetacean evidence can and cannot support in terms of language-likeness.

5 Returning to the Cetacean Evidence

The preceding sections established that while Zipf’s law of abbreviation and Menzerath’s law are solidly present in cetacean vocalizations, they are not sufficient markers of language-like communication on their own. An additional set of observable signal properties including duality of patterning, productive vocabulary growth, displacement, and context-sensitive reduction would give much stronger evidence. This section goes over what current cetacean literature has shown with respect to these.

The best candidate for displaying duality of patterning is the sperm whale. Sharma et al., 2024 found that coda types co-occur non-randomly in their exchanges, which suggests that sequence-level organization carries information to a further extent than just individual coda identity. This is suggestive, but combinatorial structure between meaningful units is not duality of patterning, which specifically requires a lower level of meaningless

units combining into meaningful ones. That lower level has been proposed by Andreas et al., 2022, individual click features could function in a similar manner to phonemes, but this has yet to be demonstrated.

Productive vocabulary growth has not been tested using Heap’s law in any cetacean species. Humpback song repertoires do change culturally over time, which suggests that the system is not rigidly closed, but whether growth follows the open-ended power law pattern characteristic of human vocabulary is unknown. A test of this could be attempted with some of the already existing datasets.

Displacement and second-mention reduction are unestablished in the literature. Both would need more detailed behavioural annotation than current datasets give us, correlating vocalizations with environmental and social context for displacement, and tracking acoustic variation across sequential exchanges for reduction. Andreas et al., 2022 identify both of these as problems that the Project CETI data collection infrastructure will eventually address.

The gap in all four criteria is real, but to a large extent reflects the current state of data and methodology rather than a conclusion about cetacean capacities. The combinatorial structure found by Sharma et al., 2024 is the most promising result of those already existing. Everything else remains to be tested.

6 Conclusion

This paper set out to ask what the presence of Zipf’s law of abbreviation and Menzerath’s law in cetacean vocalizations actually demonstrates, and what further evidence would be needed for stronger claims about language-likeness.

The two laws are valuable findings. Menzerath’s law, found in 11 of 16 cetacean species by Youngblood, 2025 shows a tendency towards hierarchical compression in vocal sequences. Zipf’s law of abbreviation, which is found in humpback and blue whales, is a more demanding result with stronger theoretical ties to the properties of linguistic systems. Both findings are significant contributions to comparative biology. But both laws also hold true as general organizational principles that occur in non-referential animal signals, music, and within biological systems. None of these systems are language. This means that finding the two laws are at best necessary conditions for language-likeness, and insufficient on their own for drawing direct comparisons to human language.

Properties that would constitute stronger evidence—duality of patterning, open vocab-

ulary growth, displacement, and context-sensitive reduction—are all in principle observable from acoustic and behavioural data without requiring claims about cetacean cognition. There is currently a lack of evidence for these properties in cetacean communication, not because they are not found, but because the data and methods to test them have not yet been systematically applied. The most promising current result is the combinatorial structure found in sperm whale codas by Sharma et al., 2024, which points toward the kind of compositional organization that would be genuinely language-like if it were to hold up under further investigation.

The presence of Zipf’s law of abbreviation and Menzerath’s law in cetacean vocalization tells us something real: these are biological systems organized for efficiency, but so are genomes, music, and protein domain architecture. Human language is also organized for efficiency, but that is not what makes it language. Finding them in whale communication is therefore better understood as evidence that cetacean vocal systems share a general property of biological organization than as evidence that they share something directly with language. The question which then arises, that current data cannot yet answer, is what lies beyond this baseline of efficiency.

Acknowledgements

I would like to thank Maddie and Harald for a fun and interesting class, which made me think about aspects of communication I had not previously considered, as well as Nikita, whose ideas pointed me towards this topic.

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